



Velu Prabhakar Kumaravel

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Place of birth: Cuddalore, India | **Nationality:** Indian | **Gender:** Male | **Phone number:**

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ABOUT ME

I am an interdisciplinary researcher at the intersection of engineering and neurocognitive science, with expertise in EEG signal processing and a strong commitment to understanding the mechanics of mind/self.

WORK EXPERIENCE

POST-DOCTORAL RESEARCH FELLOW – UNIVERSITY OF CHIETI-PESCARA – 16/03/2026 – Current – ITALY

Link : https://unich.etrasparenza.it/archivio38_sovvenzioni-e-vantaggi-economici_0_169281_48_1.html

The breathing mind: How respiratory phase-related brain oscillations alter the state of consciousness in meditation

- Co-designed the experimental protocol for investigating respiration-brain coupling during Focused Attention Meditation (FAM)
- Data analysis design and implementation (in progress)

SCIENTIFIC ASSISTANT – UNIVERSITY OF OLDENBURG AND HANSE-WISSENSCHAFTSKOLLEG – 15/02/2024 – 15/02/2026 – OLDENBURG, GERMANY

Research on mobile EEG-based feedback system for auditory perceptual learning. Funded by Hearing4all – Cluster of Excellence for Hearing Research (DFG Project ID 390895286) and Hanse-Wissenschaftskolleg.

VISITING PHD STUDENT – SWISS FEDERAL INSTITUTE OF TECHNOLOGY LAUSANNE (EPFL) – 01/02/2022 – 31/07/2022 – LAUSANNE, SWITZERLAND

Developed machine learning approaches for epileptic seizure detection within the PEDESITE project (Personalized Detection of Epileptic Seizure in the IoT Era), funded by the Swiss National Science Foundation.

EDUCATION AND TRAINING

01/11/2018 – 27/02/2023 Trento, Italy

PHD IN COGNITIVE AND BRAIN SCIENCES Fondazione Bruno Kessler / Center for Mind-Brain Sciences (CIMeC), University of Trento

Ph.D. research focused on mobile EEG solutions for studying neural correlates of cognitive functions in newborns and infants. Developed a novel MATLAB-based EEG preprocessing pipeline to enhance signal-to-noise ratio, released as open-source software to promote accessibility and reproducibility in neuroimaging research.

Final grade Cum Laude | **Thesis** Signal Processing Methods for Reliable Extraction of Neural Responses in Developmental EEG

24/09/2015 – 16/03/2018 Bologna, Italy

MASTER OF SCIENCE IN TELECOMMUNICATIONS ENGINEERING University of Bologna

Final grade 103/110 | **Thesis** Experimental Evaluation of BITalino: A Low-cost Modular Platform for Biosignals Acquisition

Final grade 7.98/10 | **Thesis** Biomedical Image Segmentation using Combined Watershed and Level Set Method**LANGUAGE SKILLS**Mother tongue(s): **TAMIL**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
ITALIAN	B1	A2	A2	A2	A2
GERMAN	A1	A1	A1	A1	A1

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user***PUBLICATIONS**

2022

[NEAR: An artifact removal pipeline for human newborn EEG data](#)

This paper presents NEAR, a novel automated artifact-removal pipeline for human newborn EEG data, designed to improve the quality of neonatal EEG recordings by detecting bad channels and adapting artifact subspace reconstruction to the unique noise characteristics of newborn data. NEAR was shown to outperform existing methods in reproducing established EEG responses from noisy developmental datasets. The software can be found online at <https://github.com/vpKumaravel/NEAR>.

SI: Developmental EEG Methods - A Tutorial Approach (Developmental Cognitive Neuroscience)

Authors: Velu Prabhakar Kumaravel, Elisabetta Farella, Eugenio Parise, Marco Buiatti

2022

[Adaptable and Robust EEG Bad Channel Detection Using Local Outlier Factor \(LOF\)](#)

This paper proposes a robust EEG bad channel detection method based on the Local Outlier Factor (LOF) algorithm, which adapts to different EEG datasets by identifying noisy channels relative to local clusters rather than global distributions. The approach was validated across newborn, infant, and adult EEG data and shown to outperform existing state-of-the-art methods in detecting bad channels.

Sensors, MDPI Journal

Authors: Velu Prabhakar Kumaravel, Marco Buiatti, Eugenio Parise and Elisabetta Farella | **Volume, Issue and Pages:** Biomedical Sensors

2021

[Efficient Artifact Removal from Low-Density Wearable EEG using Artifacts Subspace Reconstruction](#)

This conference paper presents an evaluation of the Artifact Subspace Reconstruction (ASR) algorithm for efficient automatic removal of transient and non-stationary artifacts from low-density wearable EEG recordings. The study demonstrates that ASR can significantly improve signal quality even with as few as eight channels in mobile EEG systems, supporting its use for real-world neural signal acquisition outside laboratory settings.

International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)

Authors: Velu Prabhakar Kumaravel Fondazione Bruno Kessler, Trento, Italy, CIMeC, University of Trento, Italy ; Victor Kartsch; Simone Benatti; Giorgio Vallortigara; Elisabetta Farella; Marco Buiatti | **Journal Name:** EMBC

2025

[Contrasting cognitive, behavioral, and physiological responses to breathwork](#)

This study evaluated the efficacy of MindGym-generated immersive anxiolytic experiences delivered via 360-degree VR, measuring physiological responses including ECG, EDA, EMG, and respiration, alongside cognitive performance and anxiety reduction. Results showed that VR delivery maintained the full effectiveness of the original MindGym experiences, supporting the translation of immersive therapeutic content to accessible platforms.

My role in this project was to develop preprocessing algorithms for ECG, EDA, and respiration signals, and to perform quality control analysis on the preprocessed data to include or exclude subjects based on signal quality.

Authors: Ninette S, Micah J, Caitlin L, Geena W, Velu K, Taylor K, Félix S, Nicco R | **Journal Name:** PLOS Mental Health

● HONOURS AND AWARDS

09/01/2026

Finalist of Aldo Fasolo Science Communication in Neuroscience Award – INTO Brain (Turin, Italy)

Link <https://www.intobrain.it/en/news/premio-aldo-fasolo-i-finalisti-2025/>

30/09/2022

Think Open @ CIMEC Awards 2022 – Center for Mind/Brain Sciences, University of Trento, Italy

Received recognition for the open-source toolbox NEAR, designed following FAIR principles and contributing significant scientific value to the field of developmental cognitive science.

Link https://twitter.com/cimec_unitrento/status/1575797402061049857

28/01/2021

CIMEC Doctoral Day - Best Poster Award – Center for Mind/Brain Sciences, University of Trento, Italy

Best Poster Award for presenting my research on the NEAR toolbox at CIMEC Doctoral Day.

Link https://github.com/vpKumaravel/dsday2021_poster

Star Award Winner for Two Consecutive Times (2014) – Larsen & Toubro Infotech

Quarterly award recognizing top performers in the Business Unit.

● PROJECTS

Open-Source Contributions

1. **MNE - Python:** Contributed the implementation of LOF - based bad channel detection (`mne.preprocessing.find_bad_channels_lof`), now included in the official MNE - Python library for automated M/EEG preprocessing (available since v1.7).
2. **EEGLAB Plugin** - Automated bad channel detection and artifact removal for newborn and infant EEG data (NEAR pipeline), compliant with FAIR principles.

Links https://github.com/mne-tools/mne-python/blob/maint/1.7/mne/preprocessing/_lof.py | <https://github.com/vpKumaravel/NEAR>

● GRANTS

H4A & HWK Fellowship

Awarded a 24-month fellowship on “Mobile EEG-based Feedback System for Auditory Perceptual Learning,” including a 9-month research stay at Hanse-Wissenschaftskolleg and a 15-month research position at the University of Oldenburg.

Link <https://hearing4all.de/en/career/fellowships/h4a-hwk-fellowship/>

15/09/2015 – 31/07/2016

University of Bologna Action 1 Grant

Full tuition fee waiver awarded by the University of Bologna (Italy) to outstanding international students.

● SKILLS

Professional Skills

- Laboratory techniques (EEG, EDA, EMG recordings)
- Signal acquisition and preprocessing
- Data analysis and visualization
- Experimental design
- Building Machine Learning models for EEG classification

Digital and Technical Skills

- Programming languages and frameworks
- Data analysis software (MATLAB/EEGLAB, Python/MNE-Python)
- Version control / code repositories (GitHub, GitLab)
- Office tools (MS Office, Google Workspace)
- Writing and Citation tools (LaTeX, Zotero)

● ACADEMIC SERVICE

Peer Review

Reviewer for Journal of Psychophysiology, Journal of Neural Engineering, Biomedical Physics & Engineering Express, Frontiers in Neuroergonomics, Neuroscience, IEEE EMBC, IEEE COINS, DATE.

Conference Committees

Technical Program Committee Member, DATE Conference 2025 for the track "Smart Society and Digital Wellness"

● ADDITIONAL INFORMATION

Professional Interests

- Brain-body interactions
- Consciousness and Sense of Self
- Seamless neurotechnology for mental wellness

Personal hobbies

- Reading non-fiction
- Hiking and Walking
- Meditation and Pranayama